

Linear Models Lecture 17: Pooled and Panel Data

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From Cross-Sections to Repeated Observations

- A single cross-section identifies effects only by comparing units. Adding time lets us compare a unit to itself.
- Time can enter the model in several ways:
 - As an indicator (before/after a policy) or as a continuous trend.
 - Through interactions, allowing slopes to vary by period (the effect of a free fedora on employment may differ today from 1940).
- The simplest design exploits two groups at two times: **difference-in-differences**. We treat it as the on-ramp to panel data.

Differences in Differences

Voter ID law in Texas; Oklahoma is the control. Turnout Y observed before and after.

Parameters: M = Texas baseline (e.g. borders Mexico), O = Oklahoma baseline;
 T = common period effect (e.g. Trump on the ballot), D = effect of the voter ID law.

Cross-section (after)

Texas	$M + T + D$
Oklahoma	$O + T$
Diff	$(M - O) + D$

Time-series (Texas)

Before	M
After	$M + T + D$
Diff	$T + D$

Neither comparison alone recovers D .

Difference-in-Differences: Recovering D

Difference each state across time, then difference the within-state differences.

Difference-in-Differences

	Before	After	After – Before
Texas	M	$M + T + D$	$T + D$
Oklahoma	O	$O + T$	T
Texas – Oklahoma:			D

Within-state differences cancel M and O ; differencing them cancels T .

Potential Outcomes

To define the assumption that lets DD work, restate the setup formally. D stays the effect size; use G for the group indicator.

- $G \in \{0, 1\}$: group indicator (1 = treated group, e.g. Texas; 0 = control, e.g. Oklahoma).
- $t \in \{0, 1\}$: period (0 = before, 1 = after).
- Y_t^g : potential outcome in period t if group status were g .
- Observed outcome: $Y_t = G Y_t^1 + (1 - G) Y_t^0$.
- Two are observed, two are counterfactual:
 - $Y_t^1 \mid G = 0$: Oklahoma's outcome had it been treated.
 - $Y_t^0 \mid G = 1$: Texas's outcome had it not been treated.
- Unit-level treatment effect: $\delta_i = Y_t^1 - Y_t^0$. Across the population this averages to D .

Parallel Trends

DD recovers D only if Texas and Oklahoma would have moved together absent the law.

Parallel trends assumption

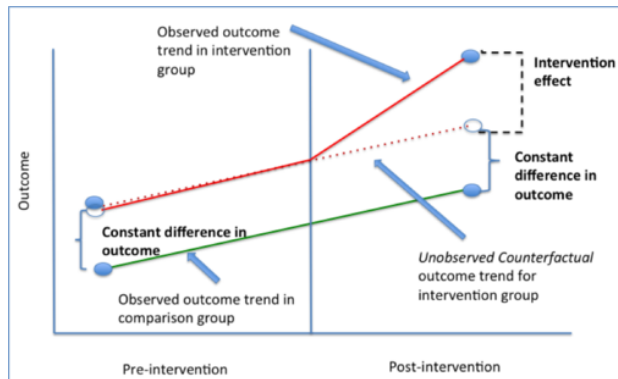
$$E[Y_{t=1}^0 \mid G = 1] - E[Y_{t=0}^0 \mid G = 1] = E[Y_{t=1}^0 \mid G = 0] - E[Y_{t=0}^0 \mid G = 0]$$

In the running example: T enters both states the same way, and the baseline gap $M - O$ does not drift over time.

Violations:

- Different state-level trends (Texas suburbanizing faster than Oklahoma).
- Treatment-induced selection or composition change (voters moving out of Texas after the law).

Parallel Trends, Visually



The dotted line is the counterfactual under parallel trends: it follows the comparison group's slope. The gap on the right is the treatment effect.

Obama Support: The Identification Problem

- Suppose we want to study the effect of racial co-identification (non-identification) on feeling thermometers of Obama in 2008.

$$ObamaRating_i = b_0 + d_O Black_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- However, we are worried about the two kinds of omitted factors:
 - Time trends (T): Obama ran after a financial crisis and war that might make him popular generally.
 - Group Differences (M-O): Black and white people might differ in their preferences over any president that is a relatively liberal northern antiwar Democratic senator.
- Suppose Obama's appeal (repulsion) among Black (white) people depends on the economic conditions in 2008, his being a liberal Senator, and not only his race?

Obama Support: DiD with a Comparison Candidate

- We can use a repeated cross section to difference out similar attitudes toward another Democratic candidate.

$$Kerry_i = b_0 + d_K Black_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- If we assume that
 - All time varying factors other than the candidate's race affected Black and white people's response to Democratic nominees equally.
 - All differences in evaluations of D. nominees between Black and white people did not change between the Kerry and Obama years.
- Then δ is an estimate of the Average Treatment Effect of the candidate's race on attitudes.

$$\delta = d_O - d_K = (\bar{r}_{O,B} - \bar{r}_{O,W}) - (\bar{r}_{K,B} - \bar{r}_{K,W})$$

Estimating Differences in Differences

- Stack the Kerry and Obama waves. Define $yr08_i = 1$ for 2008 survey takers and $B_i = \mathbb{I}(\text{Race} = \text{"Black"})$.
- **Unrestricted** (year-specific race effect):

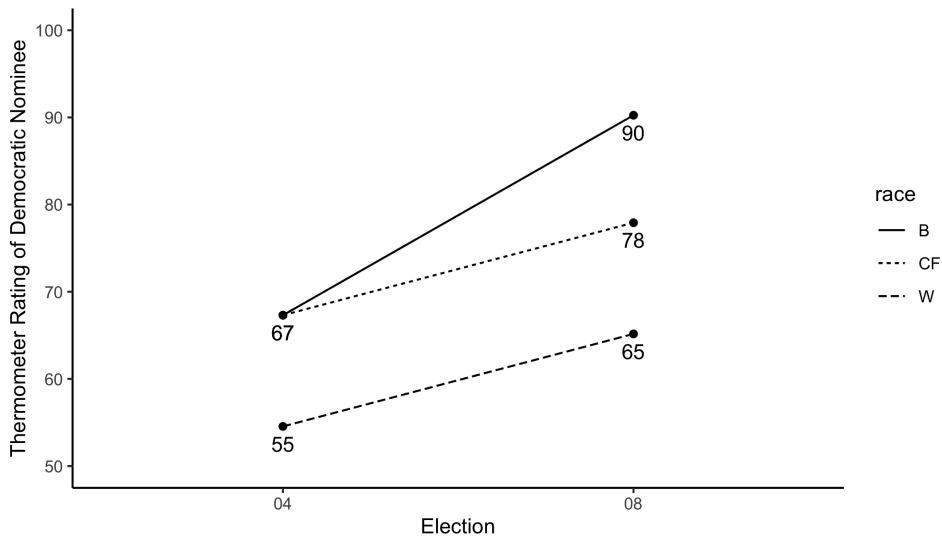
$$DRating_i = b_{0,O} yr08_i + b_{0,K} (1 - yr08_i) + d_O yr08_i B_i + d_K (1 - yr08_i) B_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- **Restricted** ($d_O = d_K \equiv d$):

$$DRating_i = b_{0,O} yr08_i + b_{0,K} (1 - yr08_i) + d B_i + \mathbf{b}'_x \mathbf{x}_i + e_i$$

- The F test of $H_0: d_O = d_K$ comparing the two models is the DiD test: the racial co-identification effect is $d_O - d_K$.

	demft		
l(1 - y08)	54.5*** (1.4)	53.0*** (1.4)	
y08	65.2*** (1.3)	65.9*** (1.3)	10.6*** (1.0)
black		21.7*** (1.0)	12.8*** (1.9)
l(1 - y08):black	12.8*** (1.9)		
y08:black	25.1*** (1.2)		12.3*** (2.3)
age	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)
Constant			54.5*** (1.4)



From Two Cells to Many: Panel Data

- DiD compared two groups at two times. Differencing cancelled time-invariant group features and group-invariant time trends.
- Panel data observes the *same* units repeatedly, so we can perform that cancellation for every unit, every period.
- Two costs of relaxing the two-cell setup:
 - Composition: groups must remain comparable across periods (no entry or exit driven by treatment).
 - Dependence: repeated observations on the same unit are correlated, breaking iid.
- The remaining sections build pooled, between, and within estimators on this richer structure.

Three Examples to Keep in Mind

- Each pairs a time-varying outcome Y_t , a time-varying regressor Z_t , and an unobserved unit-level feature A that we cannot measure directly.
- The same machinery (within transformation, GMM moment conditions) applies to all three, but the assumptions differ.
- As we proceed, ask of each example: which estimator handles it, and which would miss?

Panel Example 1: Voting behavior

- Consider a population of families. Choose one at random.
- For each family member t , we observe voting decisions Y_t and exposure to a persuasive messages Z_t .
- Each family has a household wealth W and a set of stories passed down from their grandparents A .
- We observe Y_t , Z_t and W , but not A .
- We want to estimate the effect of Z_t on Y_t , holding constant W and A .

Panel Example 2: Airline cost structure

- Consider a population of airlines. Choose one at random.
- In year t , the airline incurs cost Y_t to fly Z_t passengers.
- Each airline has a fixed efficiency A (technology, route network, labor contracts) that does not vary over time.
- We observe N randomly sampled airlines but do not measure A .
- Costs scale with passengers, but different airlines spend different amounts to fly the same load. We want the within-airline cost-output elasticity, separating scale economies from efficiency A .

Panel Example 3: Candidate Electoral Performance

- Consider a population of candidates. Choose one at random.
- In election cycle t , the candidate receives vote share Y_t .
- Each candidate has an underlying charisma A (intrinsic appeal, base of supporters) that does not vary over time.
- We observe N randomly sampled candidates across cycles, but do not measure A .
- Candidates with strong cycles often have strong follow-up cycles, partly because last cycle's success builds name recognition, donor networks, and incumbency benefits, and partly because they have higher charisma. We want the persistence of vote share, $Y_{t-1} \rightarrow Y_t$, separating that carryover from A .

What These Examples Motivate

- Voting in families: A (grandparents' stories) likely correlates with Z (messages); pooled OLS is biased and **fixed effects** restore comparability.
- Airline costs: A (proprietary technology) does not vary in time, so the **within** estimator removes it without measuring it.
- Candidate performance: Y_{t-1} appears on the right-hand side; the within transformation correlates with the error and **dynamic panel GMM** (next lecture) instruments with lagged levels.

Panel Data: Terminology

- The voters in a family, the airlines and candidates over time are all examples of *panel data*.
- Our observations are double indexed, usually divided into N units and T time periods/groups.
- If there are many T but few N , we call the data long and narrow, (time series cross section).
- If every unit is observed in every period/ group, the data are called balanced.

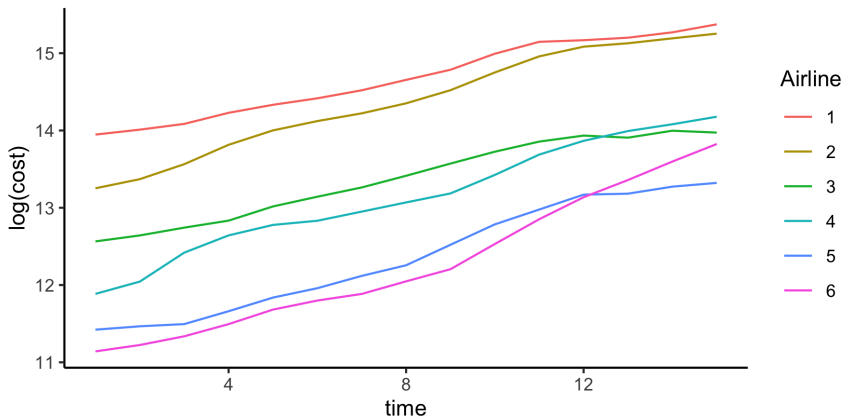
Greene's Airline Panel

- Greene analyzes the cost structure in the airline industry from 1970-84.
- Research question: do airlines have positive economies of scale?
- `library(AER); data("USAirlines")`; analyzes determinants of total costs for 6 airlines.
- The cost function follows a Cobb-Douglas specification. Taking logs yields a linear regression:

$$\ln(\text{cost}) = \beta_0 + \beta_1 \ln(\text{output}) + \beta_2 \ln(\text{fuel price}) + \beta_3 \text{load} + u$$

- *cost*: total costs of labor materials etc in USD 1000.
- *output*: revenue passenger mile index
- input *prices*: fuel prices
- *load*: share of seats sold on flights flown.

Airline Panel: Six Carriers, 15 Years



Log total cost, 1970–1984. Levels differ across carriers; all rise together over time.

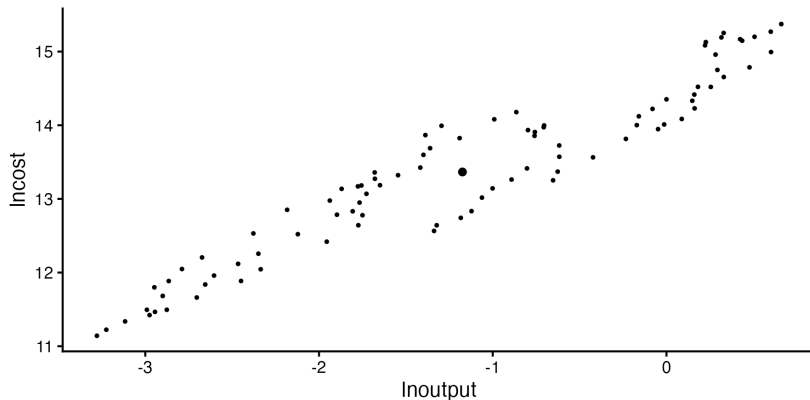
Pooled Regression of $\log(\text{costs})$

	Greene 2000	Greene 2003
(Intercept)	9.52*** (0.23)	9.42*** (0.23)
lnoutput	0.88*** (0.01)	0.94*** (0.03)
lnpfuel	0.45*** (0.02)	0.46*** (0.02)
load	-1.63*** (0.35)	-1.54*** (0.34)
lnoutput ²		0.02* (0.01)
R ²	0.99	0.99
Num. obs.	90	90

Where Does the Variation Come From?

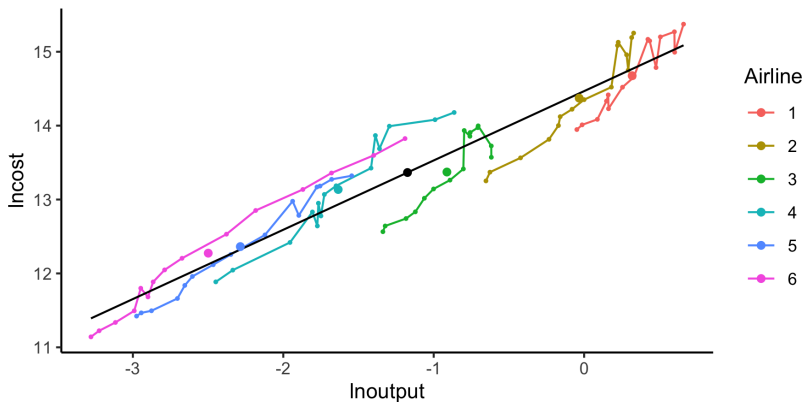
- Before reading off pooled coefficients, look at the data the way pooled OLS does: every airline-year is one point.
- Then strip off the airline-specific mean and ask what is left within an airline.
- The next plots compare those views for output, fuel price, and load.

Cost and Output: Raw Scatter



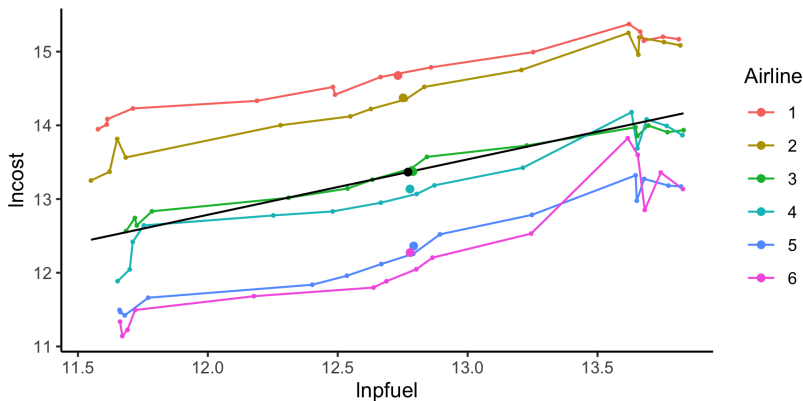
90 airline-years pooled. The black dot is the grand mean ($\overline{\ln \text{output}}$, $\overline{\ln \text{cost}}$), the only summary a single OLS fit will use.

Cost and Output: Pooled, Between, Within



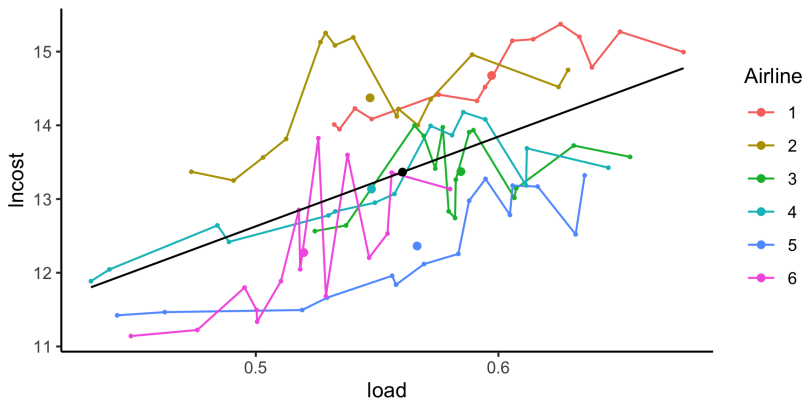
Coloured lines: each airline through time (within variation). Large coloured dots: airline means (between variation). Black line: pooled OLS fit.

Cost and Fuel Price



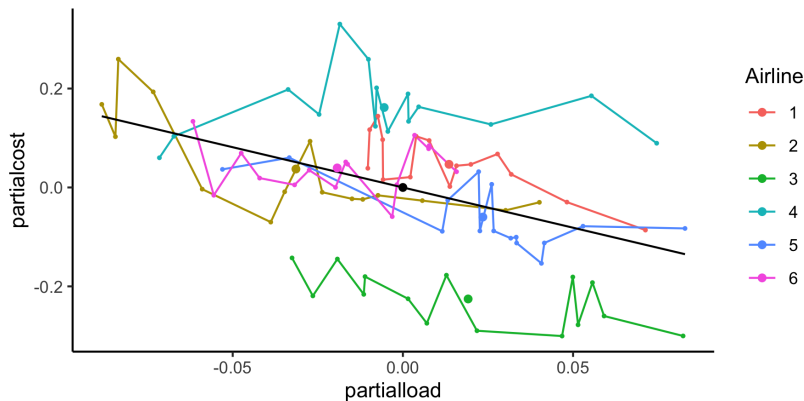
Within and between move the same way (prices rose over the 1970s for everyone), so the pooled slope is close to the within slope.

Cost and Capacity Utilization (load)



Bivariate pooled slope is mildly *positive*: airlines that run fuller planes also have higher total costs, because larger carriers do both.

Cost and Load, partialling out output and fuel price



Residual scatter from regressing both variables on (lnoutput, lnfuel). Slope flips to -1.63 : holding scale and input prices fixed, fuller planes are cheaper.

What the Plots Show

- For output and fuel price, within and between push the same way, so pooled OLS looks fine.
- For load, between-airline variation reverses the sign relative to within. The bivariate pooled scatter is misleading.
- Partialling out the other regressors recovers the within-airline relationship. FE will do this for all variables at once.

Three Flavors of Exogeneity in Panel Data

Each panel estimator needs a different version of “ $\mathbf{x}$ uncorrelated with $\mathbf{e}$ ” across time. Listed weakest to strongest:

Assumption	Formal statement	Used by
Contemporaneous	$E[e_{it} \mathbf{x}_{it}] = 0$ (plus α_j uncorrelated with $\mathbf{x}_{it}$, else FE)	Pooled OLS
Sequential (predetermined)	$E[e_{it} \mathbf{x}_{i1}, \dots, \mathbf{x}_{it}] = 0$ feedback $e_{it} \rightarrow \mathbf{x}_{i,t+1}$ allowed	Arellano-Bond (next lecture) with lagged Y as a regressor
Strict	$E[e_{it} \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}] = 0$ for all t no feedback, no anticipation	Within (FE), Between

What violates strict exogeneity. Lagged dependent variables on the right-hand side, feedback from Y to future X , anticipation of future treatment, time-varying confounders.

Autocorrelation versus Autoregression

$$Y_t = \alpha Y_{t-1} + \beta X_t + \nu_t$$

$$\nu_t = \rho \nu_{t-1} + e_t$$

- $\rho \neq 0, \alpha = 0$ has a "common factor" problem (autocorrelation):
 - Outcomes at $t - 1$ are associated with outcomes at t for many unobserved reasons.
 - OLS slope estimates are unbiased but inefficient and produces wrong standard errors.
 - Addressed with GLS/ serial correlation robust standard errors.
- $\alpha \neq 0$ is associated with dynamics:
 - Outcomes at $t - 1$ affect outcomes now.
 - OLS is biased, inefficient and produces wrong standard errors.
 - Addressed with lagged variables and other structural time series approaches.

Improvements on Inference

- In panel data we likely violate of iid errors, as observations from the same group/firm are likely associated

$$E[e_{ig}e_{jg}] = \rho\sigma_e^2 > 0$$

- Here ρ is the intra-group/firm correlation.

Standard error problems.

- If the number of groups is large, we can cluster standard errors using a formula similar to White standard errors.
- If the number of groups is small, we can just inflate our standard errors by $\sqrt{1 + (n_g - 1)\rho}$ (the Moulton factor; n_g is the (average) cluster size).
- There is a bootstrap solution that has good performance

White vs Cluster Robust

	(white)	(cluster by airline)
(Intercept)	9.52*** (0.22)	9.52*** (0.38)
lnoutput	0.88*** (0.01)	0.88*** (0.02)
lnpfuel	0.45*** (0.02)	0.45*** (0.03)
load	-1.63*** (0.32)	-1.63* (0.44)
R ²	0.99	0.99
N Clusters		6

From Fixing Inference to Using the Panel Structure

- Clustered SEs adjust the variance of $\hat{\mathbf{b}}_P$ but leave the coefficient alone.
- The deeper problem is bias when α_j is correlated with $\mathbf{X}_{it}$: pooled OLS averages it in.
- Each unit is observed T times. Differencing each variable from its own unit mean cancels α_j without ever observing it.
- That move defines the within (fixed-effects) estimator. We derive it next, then show that pooled OLS is itself a weighted average of within and between.

Fixed Effects: Two Equivalent Approaches

Panel model $y_{it} = \beta' \mathbf{x}_{it} + \alpha_i + e_{it}$, with $E[e_{it} | \alpha_i, \mathbf{x}_i] = 0$ but $E[\alpha_i | \mathbf{x}_{it}] \neq 0$ (so pooled OLS is biased).

Dummy variables

Add one dummy per unit, run OLS:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{D}\alpha + \mathbf{e}.$$

The N fitted dummies absorb the unit intercepts.

Demean within units, then regress

Subtract each variable's own unit mean and run OLS:

$$y_{it} - \bar{y}_i = \beta'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + \tilde{e}_{it}.$$

The α_i cancels because $\alpha_i - \bar{\alpha}_i = 0$.

Both produce the same $\hat{\beta}$. Why: Frisch-Waugh-Lovell turns the dummy regression into a regression on within-demeaned variables. The next three slides prove it.

Why the Two Match: Frisch-Waugh-Lovell

By FWL, the slope on $\mathbf{X}$ in the dummy regression equals OLS run on $\mathbf{X}$ and $\mathbf{y}$ after partialling $\mathbf{D}$ out of both:

$$\hat{\beta} = (\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_D\mathbf{y}, \quad \mathbf{M}_D = \mathbf{I} - \mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'.$$

$\mathbf{M}_D$ is the residual maker for the dummy regression: $\mathbf{M}_D\mathbf{v}$ is whatever is left of $\mathbf{v}$ after a regression on the unit dummies alone.

Claim. For any variable $\mathbf{v}$, $\mathbf{M}_D\mathbf{v}$ is $\mathbf{v}$ minus its unit mean: $(\mathbf{M}_D\mathbf{v})_{it} = v_{it} - \bar{v}_i$. Granted the claim, the dummy estimator runs OLS on the within-demeaned data. The next two slides verify the claim for $N = 2$.

Computing M_D for $N = 2$

- $D = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$ where i is a $(T \times 1)$ vector of 1s.

$$\begin{aligned}
 M_D &= I - D(D'D)^{-1}D' \\
 &= \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \left[\begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \right]^{-1} \begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \\
 &= \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} T & 0 \\ 0 & T \end{pmatrix}^{-1} \begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \\
 &= \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} - \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1/T & 0 \\ 0 & 1/T \end{pmatrix} \begin{pmatrix} i' & 0 \\ 0 & i' \end{pmatrix} \\
 &= \begin{pmatrix} I - \frac{1}{T} i i' & 0 \\ 0 & I - \frac{1}{T} i i' \end{pmatrix}
 \end{aligned}$$

$M_D \mathbf{y}$ Is the Within-Demeaned Vector

Writing $\bar{y}_i = \frac{1}{T} \sum_t y_{it}$ for unit i 's mean:

$$\mathbf{M}_D \mathbf{y} = \begin{pmatrix} \mathbf{I} - \frac{1}{T} \mathbf{ii}' & \mathbf{0} \\ \mathbf{0} & \mathbf{I} - \frac{1}{T} \mathbf{ii}' \end{pmatrix} \begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} y_{11} - \bar{y}_1 \\ \vdots \\ y_{1T} - \bar{y}_1 \\ y_{21} - \bar{y}_2 \\ \vdots \\ y_{2T} - \bar{y}_2 \end{pmatrix}.$$

The i th block of $\mathbf{M}_D \mathbf{y}$ is $\mathbf{y}_i$ with unit i 's own mean subtracted. The same operation applied column-by-column to $\mathbf{X}$ gives $\mathbf{M}_D \mathbf{X}$. Equivalence to the dummy regression is now confirmed.

The Within Estimator: Closed Form and Computation

Equivalence confirmed: $\mathbf{M}_D \mathbf{X}$ and $\mathbf{M}_D \mathbf{y}$ are the within-demeaned data, so the FE slope is OLS on those demeaned variables:

$$\hat{\beta}_W = (\mathbf{s}_{xx}^W)^{-1} \mathbf{s}_{xy}^W,$$

$$\mathbf{s}_{xx}^W = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)', \quad \mathbf{s}_{xy}^W = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(y_{it} - \bar{y}_i).$$

In practice. Demean every variable by its unit mean and run OLS. The residual degrees of freedom shrink from $NT - k$ to $NT - N - k$ to account for the N estimated unit means.

Stacked Matrix Notation

$\mathbf{y}_i: T \times 1, \mathbf{X}_i: T \times k, \mathbf{e}_i: T \times 1, \boldsymbol{\beta}: k \times 1$

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_N \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \vdots \\ \mathbf{X}_N \end{bmatrix} \boldsymbol{\beta} + \begin{bmatrix} \alpha_1 \mathbf{1} \\ \alpha_2 \mathbf{1} \\ \vdots \\ \alpha_N \mathbf{1} \end{bmatrix} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \vdots \\ \mathbf{e}_N \end{bmatrix}$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\alpha} \otimes \mathbf{1}_T + \mathbf{e}$$

Software Implementations

- $\text{lm}(y \sim x + f_1 + f_2 + \dots + f_n)$
- $\text{plm}(y \sim x, \text{c}(\text{"unit"}, \text{"time"}))$
- $\text{fixest}::\text{feols}(y \sim x \mid \text{fe})$
- You can also demean by time averages, but you need to correct the standard errors.

Why Demeaning Removes the Unit Effect

The FE model writes

$$y_{it} = \beta' \mathbf{x}_{it} + \alpha_i + e_{it}.$$

Subtract each variable's unit mean:

$$y_{it} - \bar{y}_i = \beta' (\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + (\alpha_i - \bar{\alpha}_i) + (e_{it} - \bar{e}_i).$$

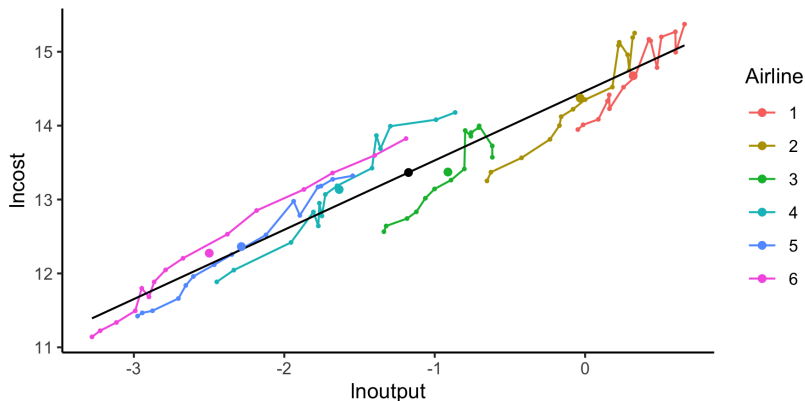
Key step. Since α_i does not vary in t ,

$$\bar{\alpha}_i = \frac{1}{T} \sum_{t=1}^T \alpha_i = \alpha_i, \quad \text{so} \quad \alpha_i - \bar{\alpha}_i = 0.$$

The unit effect drops out of the demeaned equation. Demeaning removes α_i without our ever observing or estimating it.

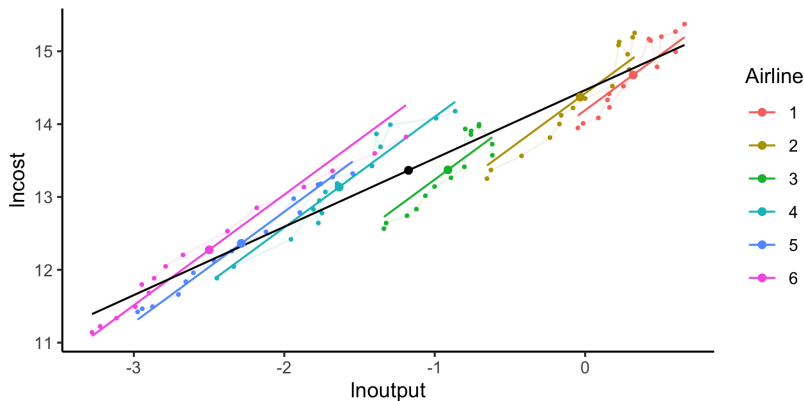
What remains, $\tilde{y}_{it} = \beta' \tilde{\mathbf{x}}_{it} + \tilde{e}_{it}$, is what OLS on the demeaned data fits.

Cost and Output: Pooled Fit



Single black line: pooled OLS slope $\hat{\beta} = 0.88$, averaging across and within airlines.

Cost and Output: Within Fit



Per-airline fits with a common slope but separate intercepts. Within slope $\hat{\beta} = 0.92$ is steeper than pooled because the flatter between-airline relationship pulls the pooled line down.

Small note on estimation

$$\mathbf{b}_{fe} = \beta + (\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}\mathbf{X}'\mathbf{M}_D\mathbf{e}$$

Under iid errors, we have: $\text{var}(\mathbf{b}_{fe}) = \sigma_e^2(\mathbf{X}'\mathbf{M}_D\mathbf{X})^{-1}$. But we need to correct our estimator for the N estimated means:

$$s_e^2 = \frac{\sum_{i=1}^N \sum_{t=1}^T e_{it}^2}{NT - N - K}$$

Fixed Effects as GMM Moment Conditions

- Define the within-transformed variables $\tilde{Y}_{it} = Y_{it} - \bar{Y}_i$ and $\tilde{\mathbf{X}}_{it} = \mathbf{X}_{it} - \bar{\mathbf{X}}_i$.
- The FE (within) estimator solves the moment conditions:

$$E[\tilde{\mathbf{X}}_{it}' \tilde{e}_{it}] = \mathbf{0}$$

- This is a **just-identified** GMM problem: k moment conditions for k parameters.
 - No overidentifying restrictions $\Rightarrow$ no J -test with FE alone.
- **Preview:**
 - Random effects adds between-group moments $\Rightarrow$ more efficient under stronger assumptions.
 - Arellano-Bond (next lecture) creates *many* moment conditions from lagged levels $\Rightarrow$ overidentification $\Rightarrow$ testable with J -statistic.

Pooled Regression Anatomy

Rewrite pooled OLS in the notation we will reuse for between and within.

- Let $\mathbf{x}_{it}$ be the $k \times 1$ vector of regressors. Grand means: $\bar{\mathbf{x}} \equiv \frac{1}{NT} \sum_{i,t} \mathbf{x}_{it}$, $\bar{y} \equiv \frac{1}{NT} \sum_{i,t} y_{it}$.
- The pooled estimator: $\mathbf{b}_P = (\mathbf{S}_{xx}^P)^{-1} \mathbf{S}_{xy}^P$ with

$$\mathbf{S}_{xx}^P = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}})(\mathbf{x}_{it} - \bar{\mathbf{x}})', \quad \mathbf{S}_{xy}^P = \sum_{i,t} (\mathbf{x}_{it} - \bar{\mathbf{x}})(y_{it} - \bar{y}).$$


```
airlines<-airlines%>% mutate(  
  mlnoutput = mean(lnoutput),  
  mlnpfuel = mean(lnpfuel),  
  mload = mean(load))  
  
Xp<-airlines%>%transmute(  
  lnoutputdm = lnoutput-mlnoutput,  
  lnpfueldm = lnpfuel-mlnpfuel,  
  loaddm = load-mload)%>%  
  as.matrix()  
  
Sp_xx = t(Xp)%*%Xp
```

Between Regression

Within threw away all cross-unit variation; **between** does the opposite: it uses only cross-unit variation.

Construction. Collapse each unit's T observations into one point at its means; run OLS on the resulting N -observation cross section:

$$\bar{y}_i = \beta' \bar{\mathbf{x}}_i + \bar{e}_i, \quad i = 1, \dots, N.$$

The slope is $\mathbf{b}_B = (\mathbf{S}_{xx}^B)^{-1} \mathbf{S}_{xy}^B$ with

$$\mathbf{S}_{xx}^B = \sum_{i=1}^N T (\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})', \quad \mathbf{S}_{xy}^B = \sum_{i=1}^N T (\bar{\mathbf{x}}_i - \bar{\bar{\mathbf{x}}})(\bar{y}_i - \bar{\bar{y}}).$$

The factor of T keeps these sums on the same scale as $\mathbf{S}^P$ and $\mathbf{S}^W$, so $\mathbf{S}^P = \mathbf{S}^B + \mathbf{S}^W$ on the next slide.

```
airlines<-airlines%>%
  group_by(airline)%>%
  mutate(
    gmlnoutput=mean(lnoutput),
    gmlnpfuel=mean(lnpfuel),
    gmload=mean(load))

Xb<-airlines%>%transmute(
  blnoutputdm = gmlnoutput-mlnoutput,
  blnpfueldm = gmlnpfuel-mlnpfuel,
  bloaddm = gmload-mload) %>%
  as.matrix()

Sb_xx = t(Xb)%*%Xb
```

```
airlines<-airlines%>%
  group_by(airline)%>%
  mutate(
    gmlnoutput=mean(lnoutput),
    gmlnpfuel=mean(lnpfuel),
    gmload=mean(load))

Xw<-airlines%>%transmute(
  plnoutputdm = lnoutput-gmlnoutput,
  plnnpfueldm = lnpfuel-gmlnpfuel,
  ploaddm = load-gmload)%>%
  as.matrix()

Sw_xx = t(Xw)%*%Xw
```

Decomposition of Variance

$$y_{it} = \bar{y}_i + (y_{it} - \bar{y}_i)$$

$$\mathbf{s}_{xx}^P = \mathbf{s}_{xx}^B + \mathbf{s}_{xx}^W$$

$$\mathbf{s}_{xy}^P = \mathbf{s}_{xy}^B + \mathbf{s}_{xy}^W$$

Pooled as a Weighted Average of Within and Between

$$\begin{aligned}
 \mathbf{b}_P &= (\mathbf{S}_{xx}^P)^{-1} \mathbf{S}_{xy}^P \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} (\mathbf{S}_{xy}^B + \mathbf{S}_{xy}^W) \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xy}^B \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^B \mathbf{b}_B \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W - \mathbf{S}_{xx}^W) \mathbf{b}_B \\
 &= (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W \mathbf{b}_W + (\mathbf{I} - (\mathbf{S}_{xx}^B + \mathbf{S}_{xx}^W)^{-1} \mathbf{S}_{xx}^W) \mathbf{b}_B \\
 &= \lambda \mathbf{b}_W + (\mathbf{I} - \lambda) \mathbf{b}_B
 \end{aligned}$$

Three estimators on the same data, differing only in which variation they use. The next slide shows them side by side on the airline panel.

Pooled, Between, and Within: Airline Estimates

	Incost		
	Pooled	Between	Within
lnoutput	0.883*** (0.013)	0.782** (0.109)	0.919*** (0.030)
lnfuel	0.454*** (0.020)	-5.524 (4.479)	0.417*** (0.015)
load	-1.628*** (0.345)	-1.751 (2.743)	-1.070*** (0.202)
Constant	9.517*** (0.229)	85.809 (56.483)	
Observations	90	6	90
R ²	0.988	0.994	0.993

What the estimates tell us

Reading the pooled column as Cobb-Douglas elasticities:

■ Output elasticity

$$\hat{\beta}_1 = 0.883 \text{ (pooled)} \implies \hat{r} = \frac{1}{\hat{\beta}_1} \approx 1.13.$$

The industry exhibits *mild increasing returns to scale* (costs rise less than proportionally with output).

■ Fuel price elasticity

$$\hat{\beta}_2 = 0.454 \implies \hat{\alpha}_F = \hat{r} \hat{\beta}_2 \approx 1.13 \times 0.454 \approx 0.51.$$

Roughly half of total variable cost is attributable to fuel.

- **Load factor effect** The coefficient on *load* (-1.63) means that a 1-percentage-point increase in capacity utilisation lowers total cost by about 1.6 %, consistent with spreading fixed operating expenses over more output.

From Cost Functions to Political Economy

- The airline panel illustrated the mechanics: removing unit heterogeneity, pooling within and between variation.
- The same machinery answers questions where the unit effect is the central concern, not a nuisance.
- Next: a country-year panel asking why top marginal tax rates rose so dramatically in the 20th century.

Political Science Application: Tax Progressivity and War

- **Scheve & Stasavage (2010, 2012):** Why did top tax rates rise dramatically in the 20th century?
- **Hypothesis:** Mass military mobilization during WWI/WWII created political pressure for progressive taxation as a “conscription of wealth.”
- **Data:** Country-year panel, 20 countries, 1850–1970.
 - Y_{it} : top marginal income tax rate
 - X_{it} : $\text{wwihighmob} \times \text{aft}$ = high-mobilization country $\times$ post-WWI
 - α_i : country fixed effects (institutions, fiscal capacity, colonial legacies)
- **Design.** A within-country DiD: did high-mobilization countries see larger jumps in top rates after the war than low-mobilization countries? Country FE absorb time-invariant differences; the identifying assumption is parallel trends across the two groups absent the war.
- **Inference.** Newey-West SEs (next slide) handle within-country serial correlation.

Scheve & Stasavage: R Implementation

```
library(plm); library(haven); library(fixest)
dta <- read_dta("progressTax.dta")
pdta <- subset(dta, year >= 1900 & year <= 1930)

# Fixed effects with plm
fe_mod <- plm(topratep ~ wwihighmobaft,
              data = pdta,
              index = c("ccode", "year"),
              model = "within")
summary(fe_mod)

# Equivalent with fixest (+ Newey-West SEs)
fe_mod2 <- feols(topratep ~ wwihighmobaft +
                 gdppcp + leftseatshp + munsuff + ratiop
                 | country,
                 data = pdta, panel.id = ~country + year,
                 vcov = NW(1) ~ country + year)
```

Where We Are: From GMM to Panel Data

- In Lectures 15–16 we developed **GMM** as a unified framework:
 - Specify moment conditions $E[\mathbf{g}(\mathbf{z}_i, \theta_0)] = \mathbf{0}$
 - Estimate $\hat{\theta}$ by minimizing $\hat{\mathbf{g}}' \mathbf{W} \hat{\mathbf{g}}$
 - Test overidentifying restrictions with the J -statistic
- **Panel data** introduces new structure:
 - Repeated observations create natural moment conditions
 - Unobserved we are dealing with heterogeneity in α_i

Panel Estimators as Moment Conditions

Estimator	Moment Condition	Key Assumption
Pooled OLS	$E[\mathbf{X}'_{it} e_{it}] = \mathbf{0}$	α_i uncorrelated with $\mathbf{X}$
Fixed Effects	$E[\tilde{\mathbf{X}}'_{it} \tilde{e}_{it}] = \mathbf{0}$	Strict exogeneity (within-unit)
Random Effects	$E[\mathbf{X}'_{it} \nu_{it}] = \mathbf{0}$	α_i uncorrelated with $\mathbf{X}$; GLS weighting
Arellano-Bond	$E[Y_{is} \cdot \Delta e_{it}] = 0, s \leq t-2$	Sequential exogeneity; lagged levels as instruments

- Each row is a **GMM estimator** with different instruments and assumptions.
- More moment conditions $\Rightarrow$ overidentification $\Rightarrow$ testable with the J -test.

When DiD Fails: Synthetic Control

- DiD recovers D only under parallel trends. For aggregate units (states, countries), one untreated average is rarely a credible counterfactual.
- Idea (Abadie & Gardeazabal 2003; Abadie, Diamond & Hainmueller 2010): match the treated unit to a *weighted average* of control units in the pre-period; the same weights deliver the post-period counterfactual.
- Athey & Imbens (2017) call it “arguably the most important innovation in the evaluation literature in the last fifteen years.”

The Weighting Problem

Setup. Treated unit $i = 0$, controls $j = 1, \dots, J$; pre-period $t \in \mathcal{T}_0$, post-period $t \in \mathcal{T}_1$.

Find weights $W = (w_1, \dots, w_J)$ with $w_j \geq 0$, $\sum_j w_j = 1$ that minimise pre-period fit error:

$$\hat{W} = \arg \min_W \sum_{t \in \mathcal{T}_0} \left(Y_{0t} - \sum_j w_j Y_{jt} \right)^2.$$

Counterfactual and effect:

$$\hat{Y}_{0t}^0 = \sum_j \hat{w}_j Y_{jt}, \quad \hat{\alpha}_{0t} = Y_{0t} - \hat{Y}_{0t}^0 \quad (t \in \mathcal{T}_1).$$

What it generalises. Linear factor model $Y_{it} = \alpha_i D_{it} + \lambda_i' \mu_t + \epsilon_{it}$: SCM lets each unit have its own multi-dimensional loading λ_i . DiD assumes a single additive unit effect; SCM handles heterogeneous responses to common factors.

Two Strong Assumptions

Convex hull

The treated unit's pre-period outcomes lie inside the convex hull of the controls' pre-period outcomes.

Perfect pre-period fit

There exist weights W such that the pre-period gap $Y_{0t} - \sum_j w_j Y_{jt}$ is zero for every $t \in \mathcal{T}_0$.

Both routinely fail. Ferman & Pinto (2021): when fit is imperfect, the SCM weights are biased and the bias does *not* vanish as $T_0 \rightarrow \infty$ if J is fixed. The mechanism is the same attenuation that plagues regression with measurement-error in regressors.

Recent Fixes

- **Powell (2018, RAND WR-1246).** Build a synthetic control for *every* unit, not just the treated. If $\hat{w}_{ij}$ puts weight 0.4 on the treated unit when matching unit i , then unit i 's post-period gap is informative about the treatment effect ($-0.4 \times$ it). Also: smooth outcomes before fitting weights to limit attenuation from idiosyncratic noise.
- **Fry (2024, J. Econometrics).** GMM-based SC: split controls into “SC units” and “instruments.” Excluded units instrument for included ones; exclusion holds when idiosyncratic shocks are uncorrelated across units. Asymptotically unbiased as $T_0 \rightarrow \infty$ with J fixed, even without perfect fit.
- Both relax the perfect-fit assumption that drives Ferman-Pinto bias.

Application: Massachusetts 2006 Health Reform

- **Question** (Powell 2018): did the 2006 Massachusetts reform (the ACA prototype) reduce mortality?
- **Why not DiD with the national average?** Massachusetts mortality does not track the national average in the pre-period, so the parallel-trend counterfactual is implausible.
- **SCM construction:** weighted average of other states fit to pre-2006 Massachusetts mortality and demographics.
- **Result:** the imperfect-SCM estimator gives a $\approx 3\%$ reduction in the Massachusetts mortality rate. A propensity-score DiD design (Sommers et al. 2014) gives comparable directional results, but the SC fit is dramatically better in the pre-period.

Summary

- Difference in difference cancels out group-invariant trends and time invariant group differences (whether of interest or not).
- Fixed effects give the same estimates as demeaning $\mathbf{X}$ and $\mathbf{y}$ by their over time means, the “within” estimator (so long as you correct the residual variance estimator).
- The within estimator corresponds to the GMM moment condition $E[\tilde{\mathbf{X}}'_{it} \tilde{e}_{it}] = \mathbf{0}$, and is just-identified.
- A “between” estimator only uses variation across units, averaged over time.
- Pooled Regression is a weighted average of the between and within estimates, with weights $(\mathbf{S}^B_{xx} + \mathbf{S}^W_{xx})^{-1} \mathbf{S}^W_{xx}$.
- **Next time:** Random effects adds between-group moments via GLS, and Arellano-Bond uses lagged levels as instruments in first-differenced dynamic panels. Both fit into the GMM framework from Lectures 15–16.